CSE 676-B: Deep Learning, Spring 2025

Assignment 0(checkpoint)

1. **Dataset Overview:**

   The dataset used is Crime Incidents in Buffalo. It contains 318735 records and 29 features. The dataset describes about different crime types, location, neighborhood and timestamp details etc.

   Main Statistics of the Dataset :

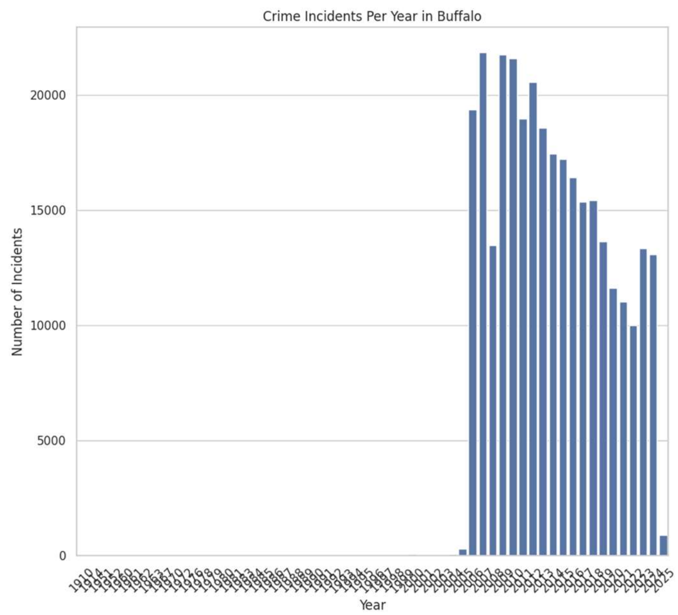
   - Total Entries: 318,735
   - Numerical Features: Latitude, Longitude, Hour of Day
   - Categorical Features: Crime Type, Day of the Week, Neighborhood.

2. **Preprocessing Techniques Used:**

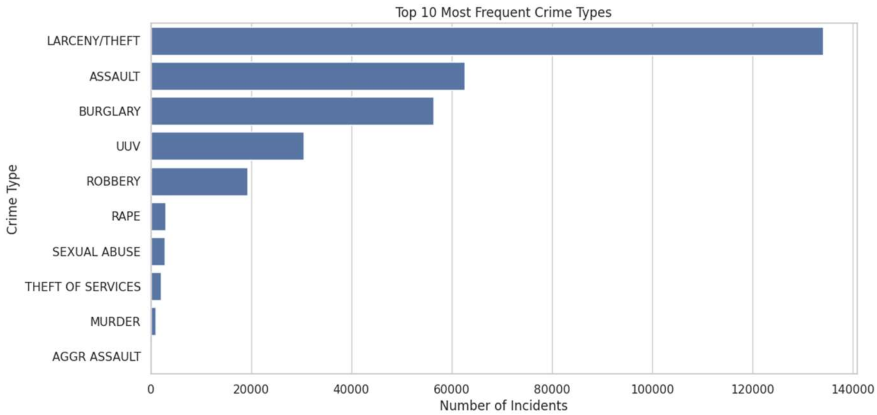
   - Dropped Irrelavant fields
   - Converted Incident Datetime to datetime format
   - Handled Missing Values
   - Encoded Categorical Feature

3. **Data Visualization & Analysis**

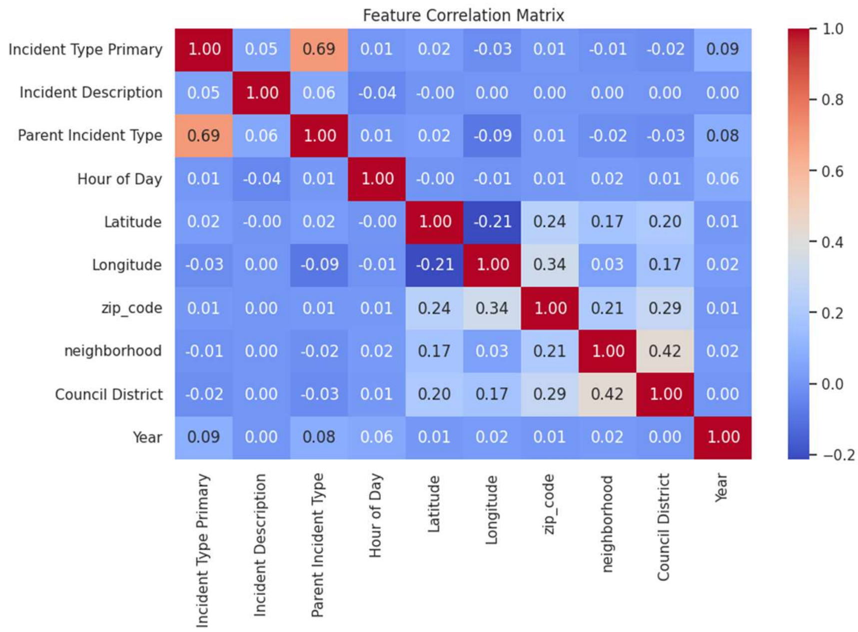
   - **Plot of Crime Incident Per Year**



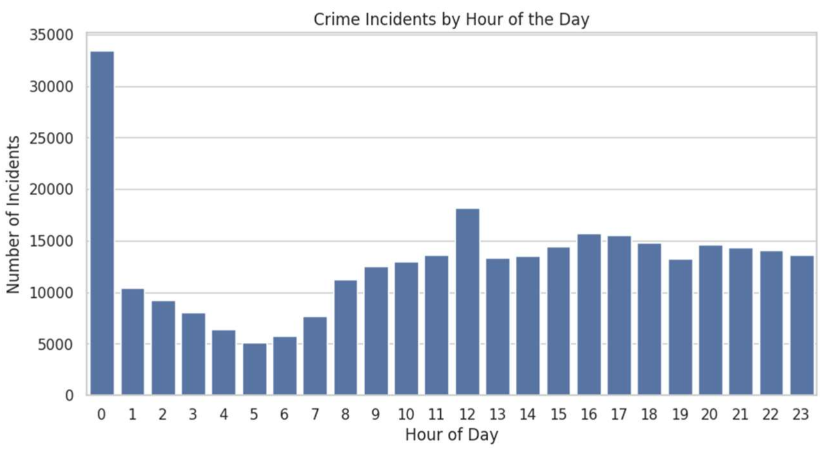
- **Plot of Top Most Frequent Crime Types**



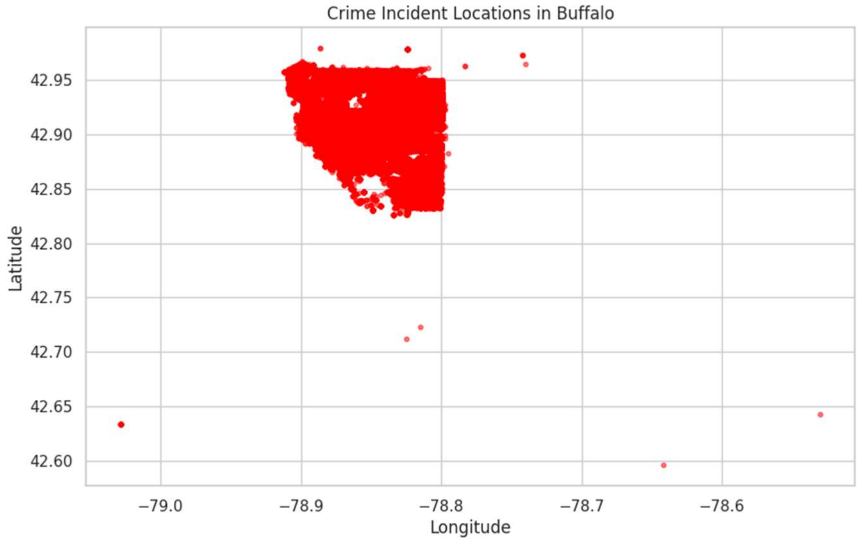
- **Correlation Matrix**



- **Plot of Crime Incidents by Hour of Data**



- **Plot of Crime Locations (Latitude vs. Longitude)**



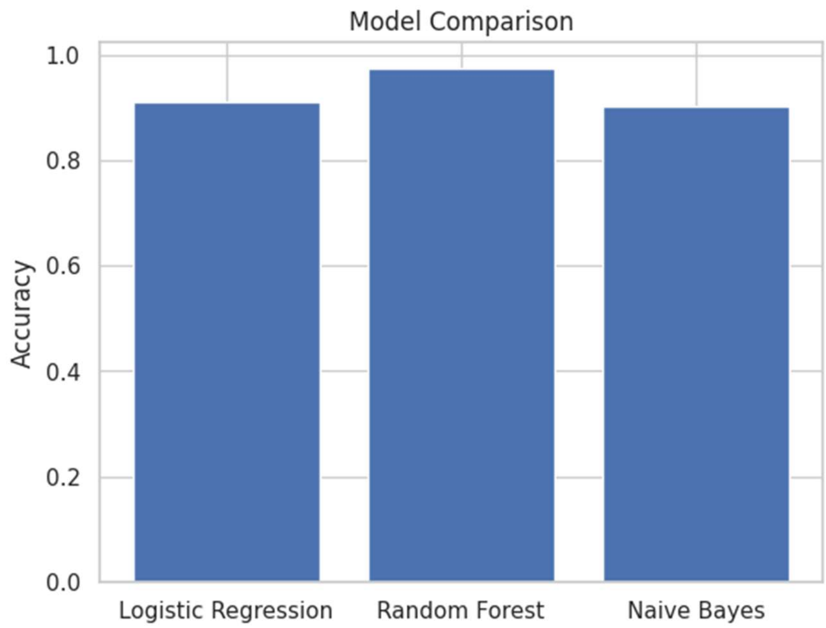
4. Machine learning Models Used:

Model	Accuracy
Logistic Regression	**91.08%**
Random Forest	**97.48**
Naïve bayes	**90.14 %**

5. NN-Model Architecture:

- Input Layer: Five Features(Hour of Day, Day of Week, Incident Type Primary, Latitude, Longitude)
- Hidden Layer : 32 neurons with ReLU activation
- Output Layer:It uses Softmax activation for classification.
- Loss Function: Cross-Entropy Loss
- Optimizer : Adam Optimizer
- Training:
- Number of Epochs=20
- Batch Size=32
- Accuracy of the Model = 87.58 %

6. Accuracy plot for the 3 ML Models



PART III OCTMNIST Classification

1. Dataset Overview

OCTMNIST dataset is a collection of 28x28 grayscale images of retinal optical coherence tomography (OCT) scans. Each image is labelled into one of 4 classes, representing different retinal diseases. Given dataset is part of the MedMNIST collection and is designed for educational purposes.

Key Statistics

1. Number of Samples:

   Training set: 10,000 images

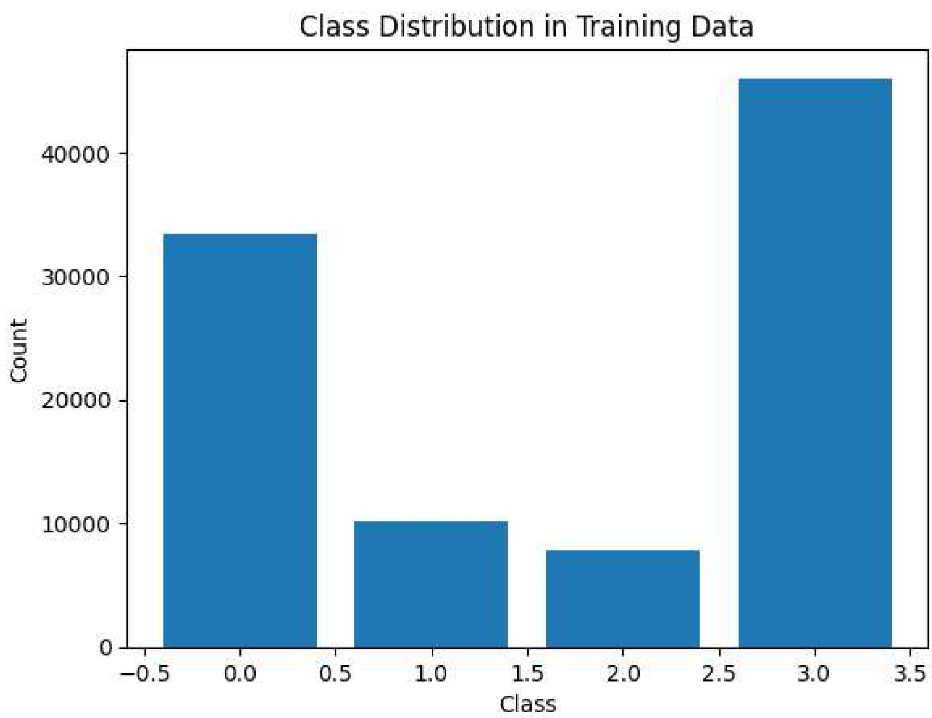
   Validation set: 2,000 images

   Test set: 2,000 images

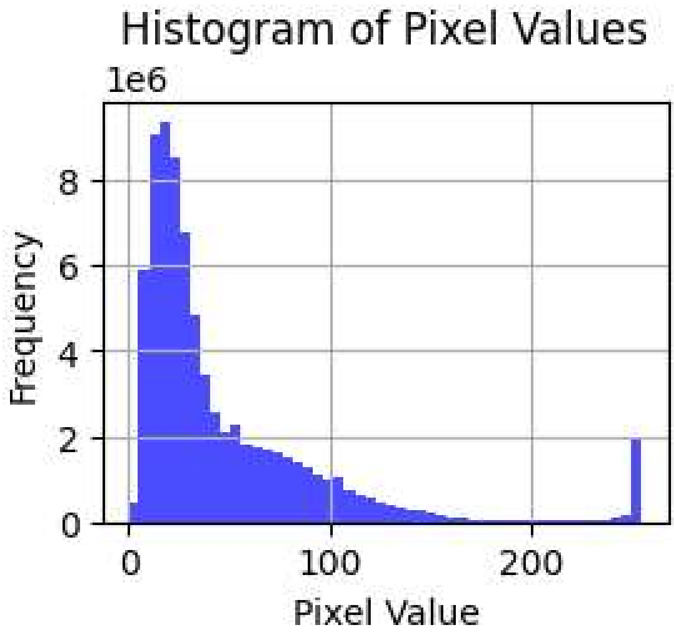
2. Number of Features: Each image has 784 pixels (28x28).
3. Mean Pixel Value: 0.123
4. Standard Deviation of Pixel Values: 0.234.
5. Missing Values: No Missing Values

2. **Visualizations**

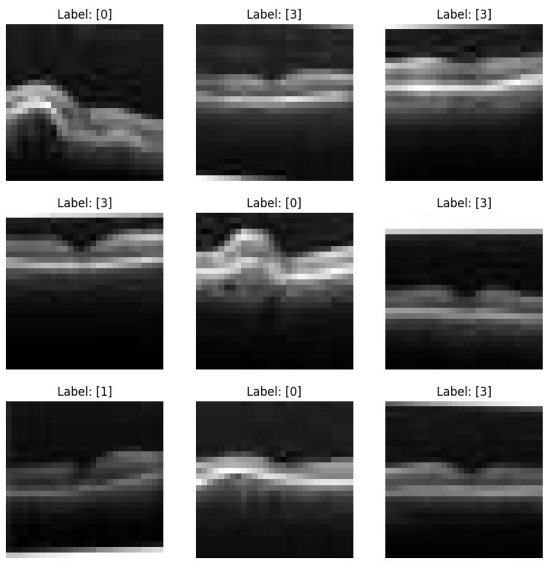
1. Class Distribution Plot



2. Histogram Plot



3 . Sample Visualizations of Data



3. Neural Network Architecture

Neural network used for this task is a **CNN** with following architecture:

- **Input Layer**: 28x28 grayscale images
- **Convolutional Layers**:

  Conv1: 32 filters, kernel size 3x3, ReLU activation, batch normalization.

  Conv2: 64 filters, kernel size 3x3, ReLU activation, batch normalization.

- **Max Pooling Layers**: 2x2 pooling after each convolutional layer.
- **Fully Connected Layers**:

  FC1: 128 neurons, ReLU activation, batch normalization, dropout (0.5).

  FC2: 4 neurons (output layer, one for each class).
- **Output Layer**: Softmax activation is used for multi-class classification.

4. Improvement Techniques Applied are:

Dropout: To avoid overfitting, dropout was implemented following the first completely connected layer. The reduced discrepancy between training and validation accuracy shows that this enhanced the model's generalization.

Early Stopping: The validation loss was tracked using early halting. If the validation loss did not decrease after five epochs, training was terminated. Training time was saved and overfitting was avoided.

Batch normalizing: Following each convolutional and fully connected layer, batch normalizing was implemented. This increased convergence speed and stabilized training.

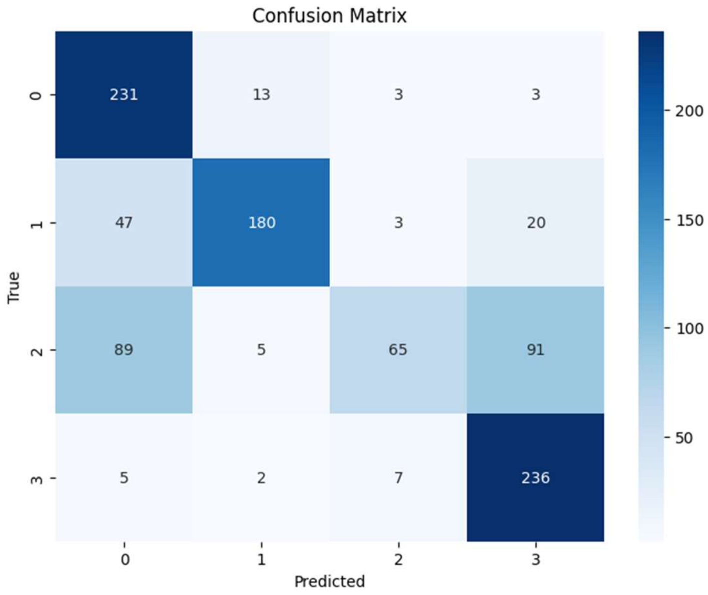
5. Results and Analysis

Training and Validation Metrices

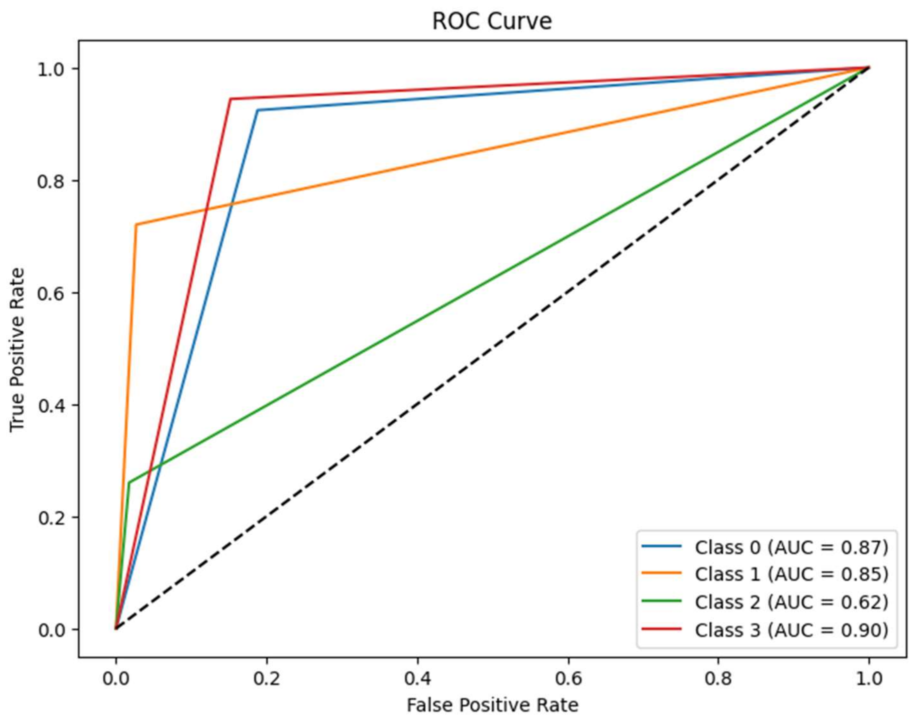
```
Epoch [1/10], Train Loss: 0.10253262293483485, Val Loss: 84.43748228354532, Train Acc: 0.9651199770202201, Val Acc: 0.7152236356175625
Epoch [2/10], Train Loss: 0.09611944747438503, Val Loss: 85.82378012235047, Train Acc: 0.9673153667018887, Val Acc: 0.7336889618383258
Epoch [3/10], Train Loss: 0.09237028728361459, Val Loss: 99.7830798665031, Train Acc: 0.9687310852816562, Val Acc: 0.740716044316783
Epoch [4/10], Train Loss: 0.08742220393012906, Val Loss: 102.44097496407932, Train Acc: 0.9706289688849677, Val Acc: 0.687320475995076
Epoch [5/10], Train Loss: 0.08325599037841686, Val Loss: 100.6108913734311, Train Acc: 0.9722498640704987, Val Acc: 0.7410750923266312
Epoch [6/10], Train Loss: 0.07991976536849509, Val Loss: 91.99278137332104, Train Acc: 0.9733373000810448, Val Acc: 0.7268157570783751
Epoch [7/10], Train Loss: 0.0765717332856363, Val Loss: 102.55840978153417, Train Acc: 0.9733988530627737, Val Acc: 0.7253795650389824
Epoch [8/10], Train Loss: 0.07184735604004538, Val Loss: 79.75535772354877, Train Acc: 0.975994337125681, Val Acc: 0.7277390233894132
Epoch [9/10], Train Loss: 0.06858916964969815, Val Loss: 89.44872922428318, Train Acc: 0.9776562676323646, Val Acc: 0.7571809601969635
Epoch [10/10], Train Loss: 0.06887243801496894, Val Loss: 102.91743535526463, Train Acc: 0.977317726232855, Val Acc: 0.7298933114485022
```

Plots of the Training vs Validation accuracy and Training Loss vs Validation Loss

Confusion Matrix



Roc Curve: The ROC curve shows high AUC values for all classes, indicating strong classification performance.



Part IV - Deep Learning Theoritical part

1. Forward - backward Pass

Given: Hidden layer activation function : ReLU

Output layer activation function : Linear

Learning Rate : 0.03

Target $y = 0.5$

1. Perform forward pass & Estimate the predicted output ($\hat{y}$)

Calculating input to each neuron in hidden layer

$h_1 = 0.7 \times w_1 + 0.5 \times w_3 + 1 \times w_5$

$= 0.7 \times (-0.3) + 0.5(0.15) + 1(0.9) = 0.765$

$h_2 = w_2 \times 0.7 + 0.5 \times w_4 + 1 \times w_6 = 0.52$

Applying ReLU Activation function to h_1 & h_2

$h_{1R} = max(0, h_1) = max(0, 0.765) = 0.765$

$h_{2R} = max(0, h_2) = max(0, 0.52) = 0.52$

h_3 is already given as 1.

$\therefore$ Final predicted output $\hat{y} = h_1R \times w_7 + h_{2R} \times w_8 + 1 \times w_9$

$\hat{y} = 0.765 \times 0.7 + 0.52 \times 0.25 + 1 \times (-0.1)$

$\hat{y} = 0.5655$

2. Estimate the MSE

Mean Squared Error (MSE) = $\frac{1}{2}(y-\hat{y})^2$

$MSE = \frac{1}{2}(0.5-0.5655)^2$

$\underline{MSE = 0.002145}$

3. Find the gradient using back-propagation

→ Gradients for output layer weights.

$\frac{\partial Error}{\partial w_7} = -h_{1R}(y-\hat{y}) = h_{1R} \cdot \frac{\partial Error}{\partial \hat{y}}$

$= -0.765(-0.0655)$

$\frac{\partial Error}{\partial w_7} = 0.0501$

$\frac{\partial Error}{\partial w_8} = -h_{2R}(y-\hat{y})$

$= 0.52 \times 0.0655$

$\frac{\partial Error}{\partial w_8} = 0.03406$

$\frac{\partial Error}{\partial w_9} = 1 \times 0.0655$

$\frac{\partial Error}{\partial w_9} = 0.0655$

3. Gradients for hidden Layer Neurons..

$$\frac{\partial Error}{\partial h_{1R}} = \frac{\partial Error}{\partial \hat{y}} \times w_7 = 0.0655 \times 0.7 = \underline{0.04585}$$

$$\frac{\partial Error}{\partial h_{2R}} = \frac{\partial Error}{\partial \hat{y}} \times w_8 = 0.0655 \times 0.25 = \underline{0.01637}$$

Since ReLU derivative is 1 for positive values

$$\frac{\partial Error}{\partial w_5} = \frac{\partial Error}{\partial h_1} = \frac{\partial Error}{\partial h_{1R}} = \underline{0.04585}$$

$$\frac{\partial Error}{\partial w_6} = \frac{\partial Error}{\partial h_2} = \frac{\partial Error}{\partial h_{2R}} = \underline{0.016375}$$

Gradients for hidden layer weights

$$\frac{\partial Error}{\partial w_1} = \frac{\partial Error}{\partial h_1} \times 0.7 = 0.04585 \times 0.7 = \underline{0.032}$$

$$\frac{\partial Error}{\partial w_3} = \frac{\partial Error}{\partial h_1} \times 0.5 = 0.04585 \times 0.5 = \underline{0.0229}$$

$$\frac{\partial Error}{\partial w_2} = \frac{\partial Error}{\partial h_2} \times 0.7 = 0.016375 \times 0.7 = \underline{0.01146}$$

$$\frac{\partial Error}{\partial w_4} = \frac{\partial Error}{\partial h_2} \times 0.5 = 0.016375 \times 0.7 = \underline{0.00818}$$

4. Update the weights.

Given learning rate $\eta = 0.03$

$$w_7' = w_7 - \eta \times \frac{\partial Error}{\partial w_7}$$

$$w_7' = 0.7 - 0.03 \times 0.0507$$

$$\boxed{w_7' = 0.698}$$

$$w_8' = w_8 - \eta \times \frac{\partial Error}{\partial w_8} = 0.25 - 0.03 \times 0.03406$$

$$w_8' = 0.2489$$

$$\boxed{w_8' = 0.2489}$$

$$w_9' = w_9 - \eta \times \frac{\partial Error}{\partial w_9} = -0.1 - 0.03 \times 0.0655$$

$$\boxed{w_9' = -0.1019}$$

$$w_1' = w_1 - \eta \times \frac{\partial Error}{\partial w_1}$$

$$w_1' = -0.3 - 0.03 \times 0.032$$

$$\boxed{w_1' = -0.30096}$$

$$w_2' = 0.8 - 0.03 \times 0.01146$$

$$\boxed{w_2' = 0.799}$$

$w_3' = w_3 - 0.03 \times 0.0229$

$\boxed{w_3' = 0.1493}$

$w_4' = w_4 - 0.03 \times 0.00818$

$\boxed{w_4' = 0.1997}$

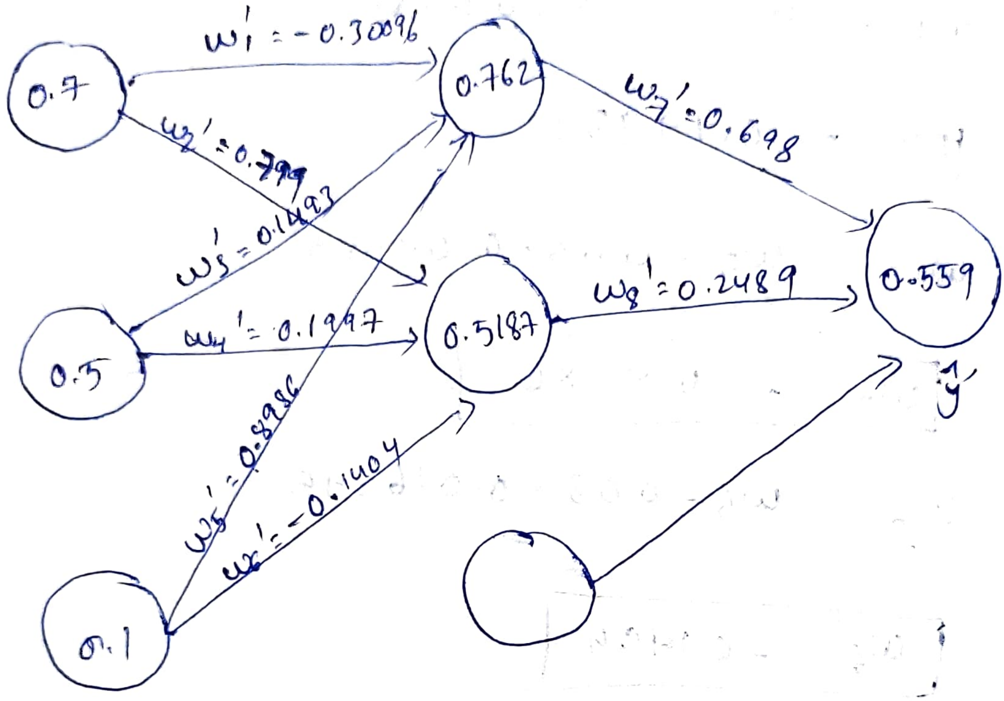
$w_5' = w_5 - 0.03 \times 0.04585$

$\boxed{w_5' = 0.8986}$

$w_6' = w_6 - 0.03 \times 0.016375$

$\boxed{w_6' = -0.1404}$

5. Draw a computation graph for the forward and backward pass



(6) Perform a forward pass to estimate the predicted output using the update weights.

$h_1' = 0.7 \times w_1' + 0.5 \times w_3' + 1 \times w_5'$

$h_1' = 0.7 \times (-0.30096) + 0.5 (0.1493) + 1 \times 0.8966$

$= 0.762$

$h_2' = 0.7 \times w_2' + 0.5 \times w_4' + 1 \times w_6'$

$h_2' = 0.5187$

$h_{1R}' = \max(0, h_1') = 0.762$

$h_{2R}' = \max(0, h_2') = 0.5187$

$h_3 = 1$

predicted output using the update weights

$\hat{y}' = 0.762 \times w_7' + 0.5187 \times w_8' + 1 \times w_9'$

$\hat{y}' = 0.762 \times 0.898 + 0.5187 \times 0.2489$

$+ 1 \times (-0.1019)$

$\boxed{\hat{y}' = 0.559}$

7. Estimate MSE & compare the results

$$MSE = \frac{1}{2}(y - y')^2$$

$$= \frac{1}{2}(0.5 - 0.559)^2$$

$$MSE = 0.0017405$$

~~comparing the MSE of the updated weights output~~

MSE in step 2 = 0.002145

After comparing the step 7 results with step 2 results, we can observe that the MSE value is reduced after applying backpropagation.

2) Derivation of Tanh

Given $f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

Required to prove :- $f'(x) = 1 - f(x)^2$

proof :- $f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Applying differentiation on both sides.

$f'(x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)$

$f'(x) = \frac{(e^x + e^{-x})\frac{d}{dx}(e^x - e^{-x}) - (e^x - e^{-x})\frac{d}{dx}(e^x + e^{-x})}{(e^x + e^{-x})^2}$

(∵ using quotient rule)

$f'(x) = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$

$f'(x) = \frac{(e^x)^2 + 1 + (e^{-x})^2 + 1 - [(e^x)^2 - 2 + (e^{-x})^2]}{(e^x + e^{-x})^2}$

$f'(x) = \frac{2 + 2}{(e^x + e^{-x})^2}$

$f'(x) = \frac{4}{(e^x + e^{-x})^2}$

$f'(x) = \frac{4}{(e^x+e^{-x})^2}$ — equation 1

from $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Squaring on both sides

$f(x)^2 = \frac{(e^x - e^{-x})^2}{(e^x+e^{-x})^2}$

$f(x)^2 = \frac{e^{2x}+e^{-2x}-2}{(e^x+e^{-x})^2}$

Add and subtract 2 in numerator.

$\therefore f(x)^2 = \frac{e^{2x}+e^{-2x}+2-4}{(e^x+e^{-x})^2}$

$f(x)^2 = \frac{e^{2x}+e^{-2x}+2}{(e^x+e^{-x})^2} - \frac{4}{(e^x+e^{-x})^2}$

Substitute equation 1 in above equation

$f(x)^2 = \frac{e^{2x}+e^{-2x}+2}{(e^x+e^{-x})^2} - f'(x)$

$f'(x) = \frac{e^{2x}+\cancel{e^{-2x}}+2}{e^{2x}+e^{-2x}+2} - f(x)^2$

$\boxed{f'(x) = 1 - f(x)^2} = 1-(\tanh(x))^2$

$\Rightarrow$ Hence, proved that $f'(x) = 1 - f(x)^2$